

Graphical Abstract

Scatter-Regularized Cross-Multi-Kernel Contrastive Learning for One-Class Anomaly Detection

Highlights

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- A cross-multi-kernel contrastive encoder reconciles multi-scale kernel views by treating the kernel index as the contrastive view.
- A multi-kernel scatter regularizer is proved equal to the average within-class scatter and to one-class centroid multiple-kernel k -means.
- A dual-signal detector fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, staying robust across datasets where either single signal collapses.

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Abstract

One-class anomaly detection has attracted extensive research due to its broad application in identifying unforeseeable deviations. The idea of contrastive learning is one of the widely used methods in modern outlier analysis. Nevertheless, these methods mainly use cross-view consistency to model normal data, so they cannot effectively guarantee the absolute intra-class compactness crucial for identifying outliers. Moreover, they typically map data into a single, fixed geometric space and cannot effectively capture complex similarity structures across multi-scale data. In this work, we propose a Scatter-regularized Cross-Multi-Kernel (SCMK) contrastive framework for one-class anomaly detection. First, a bank of multi-scale Gaussian kernels is constructed to inject multi-resolution geometric priors. Second, a cross-kernel InfoNCE objective is defined to treat the kernel index as a distinct view and align the representations. Then, a novel multi-kernel scatter regularizer is introduced to explicitly compact the normal class within every kernel-induced space. Based on this idea, two complementary signals—an angular detector (linear OC-SVM) and a radial detector (RBF OC-SVM)—are fused via a per-sample maximum to quantify the anomaly scores. Finally, experimental comparisons with state-of-the-art methods on 20 benchmark datasets show that the proposed method has a significant improvement, achieving the highest mean AUC, and applies robustly to diverse dataset topologies.

Keywords: anomaly detection, multiple kernel learning, contrastive learning, one-class classification, representation learning

1. Introduction

Anomaly detection is a fundamental task in data mining and pattern recognition, with broad applications in industrial fault diagnosis [?], network intrusion detection [?], healthcare monitoring [?], financial risk control, and cyber-physical system surveillance. Its goal is to identify observations that deviate significantly from the predominant normal patterns. This problem is intrinsically challenging because anomalies are typically rare, heterogeneous, and difficult to enumerate in advance. Existing approaches can be broadly categorized into statistical, proximity-based, reconstruction-based, and boundary-based methods. Statistical models identify samples that deviate from an assumed data distribution, whereas proximity-based detectors rely on distance or density estimates to expose isolated observations. Although these methods are effective for low-dimensional or well-structured data, their performance often degrades when the normal class exhibits nonlinear, heterogeneous, or multi-scale dependencies. Deep reconstruction models, such as autoencoders, enhance representation capacity through nonlinear mappings, but their anomaly scores are typically derived from reconstruction errors rather than from an explicit geometric description of the normal region. Consequently, they may reconstruct anomalies with small errors and provide limited control over the compactness or boundary of normal data. Kernel-based one-class methods offer a complementary route by modeling nonlinear similarities while preserving explicit geometric detection objectives in the induced feature space.

Kernel-based one-class methods are particularly appealing when anomalies are unavailable or unreliable during training. Representative methods such as OC-SVM and SVDD learn a compact description of normal data in a reproducing kernel Hilbert space: OC-SVM separates normal samples from the origin by a maximum-margin hyperplane, whereas SVDD encloses them within a minimum-volume hypersphere. These formulations provide explicit geometric detection objectives and can model nonlinear normal regions through the kernel trick. Their strength, however, also reveals a central limitation: the learned normal region is entirely determined by a preselected kernel geometry. In particular, a Gaussian kernel with a small bandwidth emphasizes local neighborhood relations, while a large bandwidth captures broader distributional

structure; committing to a single bandwidth may therefore miss complementary similarity information. Moreover, classical kernel detectors operate on fixed kernels or raw features and do not learn representations adapted to the downstream one-class objective. These limitations motivate a framework that combines multi-scale similarity information with learnable representation modeling for one-class anomaly detection.

To alleviate the limitation of a single kernel space, existing multi-kernel anomaly detection methods usually combine multiple kernels to characterize complementary similarity information. They further learn kernel weights adaptively, so as to enhance the expressive capacity of the combined kernel space for complex data structures. These methods improve the adaptability of kernel models to different data distributions and similarity scales to some extent. However, their core focus still lies in the selection, weighting, and fusion of predefined kernels. In other words, they aim to construct a more suitable combined kernel space for the downstream one-class detection objective. Although such strategies can exploit complementary information from different kernel scales, they usually do not explicitly optimize whether the same sample forms consistent representations across these kernels. Therefore, traditional multi-kernel learning alleviates the sensitivity to single-kernel selection, but still lacks a representation learning mechanism that directly models cross-kernel consistency.

Contrastive learning provides a natural tool to address this issue. It explicitly aligns the representations of the same sample under different views by constructing positive and negative pairs. Inspired by contrastive learning, we regard kernel functions at different scales as natural views. We then propose a multi-kernel contrastive learning scheme without explicit kernel-combination weights. Specifically, instead of explicitly constructing multiple RKHS representations and learning their weighted combination, we assign an independent linear projection head to each kernel scale. In this way, raw samples are first mapped into multiple same-dimensional kernel-associated embeddings. Kernel similarities are then computed on these embeddings using the corresponding kernel functions. The projection heads are optimized through a cross-kernel InfoNCE objective. Accordingly, embeddings of the same sample from different kernel-associated branches form positive pairs, whereas embeddings of different samples serve as negative pairs. Multi-kernel information is therefore no longer integrated

merely through explicit weighted fusion. Instead, the cross-kernel contrastive objective optimizes the kernel-associated projection heads and implicitly forms a cross-scale consistent multi-kernel representation space.

Although cross-kernel contrastive learning can align the representations of the same sample across different kernel scales, such cross-view consistency is not equivalent to the normal-class compactness required by one-class anomaly detection. Specifically, the contrastive objective mainly encourages different views of the same sample to be close to each other. It also maintains instance-level discriminability through negative pairs. Therefore, even if each normal sample is highly consistent across kernel views, all normal samples may still be widely scattered in the representation space. For one-class anomaly detection, such dispersion can weaken the ability of subsequent detectors, such as OC-SVM or SVDD, to characterize the normal region. This is because one-class detectors typically rely on normal samples forming a compact region with a clear boundary. To bridge this gap, we further introduce a multi-kernel scatter regularizer. It explicitly compacts the intra-class distribution of normal samples by maximizing the norm of the normal embedding centroid in each kernel view. This regularizer enables the model not only to learn cross-kernel consistent representations of the same sample, but also to encourage all normal samples to form a more compact structure in each kernel-associated space. As a result, the learned representation becomes more suitable for downstream one-class detection.

Based on the above analysis, we propose a Scatter-regularized Cross-Multi-Kernel contrastive learning framework, termed SCMK. It is designed to address the limitations of a single kernel scale, insufficient cross-kernel view consistency, and the lack of normal-class compactness. SCMK first uses multi-scale Gaussian kernels to characterize the similarity information of normal samples from local neighborhoods to global distributions. Then treats different kernel scales as natural views. A lightweight multi-head linear projection network is used to generate kernel-associated embeddings, and a cross-kernel InfoNCE objective explicitly aligns the representations of the same sample under different kernel views. Unlike contrastive learning methods that rely only on cross-view consistency, SCMK further introduces a multi-kernel scatter regularizer tailored to one-class anomaly detection. By compacting the intra-class distribution of

normal samples in each kernel-associated representation space, SCMK learns normal representations that are both cross-kernel consistent and intra-class compact. Finally, at the detection stage, SCMK exploits the complementarity between directional and magnitude signals to adapt to different types of anomaly patterns. The main contributions of this paper are summarized as follows:

1. A kernel-view-driven cross-multi-kernel contrastive learning framework is proposed. Gaussian kernels at different scales are treated as natural views, which avoids the reliance on random augmentations for view construction in conventional contrastive learning. A cross-kernel InfoNCE objective is designed to align the representations of the same sample across different kernel views, enabling consistent multi-scale representation learning.
2. To characterize multi-scale similarity information, we construct a Gaussian kernel bank calibrated by the median pairwise distance of normal samples. This kernel bank captures similarity information from local neighborhoods to global distributions. Meanwhile, a lightweight multi-head linear projection network, termed CMKNet, is adopted, where each kernel corresponds to a lightweight projection head. This design learns kernel-associated representations while keeping the model complexity low.
3. For one-class anomaly detection, a consistency-compactness learning objective and a dual-signal detection mechanism are developed. A multi-kernel scatter regularizer is introduced to explicitly compact the normal class by maximizing the centroid norm of normal embeddings in each kernel view. This compensates for the limitation that contrastive learning emphasizes cross-view consistency but lacks intra-class compactness constraints. In addition, directional and magnitude signals are extracted for anomaly scoring. A linear OC-SVM and an RBF OC-SVM are used to model angular and radial anomalies, respectively, and their scores are fused through a per-sample maximum strategy to adapt to diverse anomaly patterns.
4. Extensive experiments are conducted on 20 benchmark datasets against multi-

ple state-of-the-art anomaly detection methods. The results show that SCMK achieves superior average detection performance and stability. Statistical significance tests verify the reliability of the improvement. Ablation studies demonstrate the contributions of cross-kernel contrastive learning, multi-kernel scatter regularization, and dual-signal fusion. Parameter sensitivity and runtime analyses further show the robustness and practical applicability of the proposed method across different data settings.

The main contributions of this work are summarised as follows.

- (1) We propose a cross-multi-kernel contrastive framework that treats the kernel index as the contrastive view and aligns the embeddings of a multi-scale kernel bank whose bandwidths are auto-calibrated by a median-distance heuristic on the normal data.
- (2) We introduce a multi-kernel scatter regularizer and *prove* that it is equivalent to the average within-class scatter and to the one-class instance of centroid multiple-kernel k -means, bridging multiple-kernel clustering theory and contrastive anomaly detection.
- (3) We provide a collapse-avoidance analysis through the alignment–uniformity lens, showing how the contrastive and scatter terms balance, and we exploit the resulting representation with a lightweight dual-signal detector that fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, recovering the discriminative information that ℓ_2 normalization alone discards.
- (4) Extensive experiments on twenty benchmark datasets against seven recent detectors demonstrate that the proposed method attains the highest mean and most stable detection accuracy among competitive baselines, and ablation studies isolate the contribution of each component.

The remainder of this paper is organized as follows. Section 2 reviews related work on one-class anomaly detection, multiple kernel learning and contrastive representation

learning. Section 3 presents the proposed SCMK method. Section 4 reports the experimental comparison, statistical tests, runtime, parameter sensitivity and ablation studies. Section 5 concludes the paper and outlines future work.

2. Related work

2.1. One-class and deep anomaly detection

Unsupervised anomaly detection has been studied for decades under a variety of paradigms. Traditional approaches can be broadly grouped into statistical, distance-based, density-based and clustering-based methods. Statistical methods fit a model to the bulk of the data and flag observations that deviate significantly from the assumed distribution [5]. Distance-based methods treat points that lie far from most others in the feature space as anomalous [6], whereas density-based methods—of which the Local Outlier Factor (LOF) [3] is the canonical example—compare the local density of a point with that of its neighbours to expose local outliers. Clustering- and partition-based methods instead group the data and regard samples that fall far from any cluster, or that are easily isolated as in the isolation forest [4], as anomalies. Although effective on low-dimensional data, these methods rest on explicit assumptions about the geometry of the normal class—a roughly uniform distribution for distance-based methods, a well-calibrated neighbourhood for density-based methods, or a fixed similarity measure—and consequently degrade on high-dimensional, heterogeneous or categorical data whose structure violates those assumptions.

A particularly influential family formalises detection as *one-class* classification: only normal data are available, and the task is to characterise the compact region they occupy so that deviations can be flagged at test time. The two canonical kernel-based formulations are the One-Class Support Vector Machine (OC-SVM) [1], which separates the normal data from the origin by a maximum-margin hyperplane in a reproducing-kernel Hilbert space, and Support Vector Data Description (SVDD) [2], which encloses the normal data within a minimum-volume hypersphere; the two objectives coincide for translation-invariant kernels. These shallow one-class detectors are effective but remain sensitive to the choice of kernel and similarity measure and

scale poorly with dimension, which motivates coupling the one-class objective with a learned representation.

Deep anomaly detection learns the representation jointly with the detector. Autoencoder and robust-autoencoder approaches [7] flag samples with large reconstruction error, and Deep SVDD [8] learns a network that maps normal data into a minimum-volume hypersphere, at the risk of trivial collapse. A complementary line is self-supervised: geometric-transformation prediction [9] and its generalisation to non-image data, GOAD [10], detect anomalies through the predictability of applied transformations.

2.2. *Multiple kernel learning*

Multiple kernel learning combines several base kernels into a single, more expressive kernel, typically by learning a convex combination of weights. Foundational formulations cast weight learning as semidefinite or gradient-based optimisation [16, 17], and localised variants allow the combination to vary across the input space [18]; see [15] for a survey. In the unsupervised setting, multiple-kernel k -means and its centroid (CMKKM) and matrix-regularised variants [19, 20] jointly optimise cluster assignments and kernel weights, with diversity-promoting regularisers discouraging redundant kernels. These techniques are, however, shallow—they act on raw features and output only kernel weights. Our work retains the multiple-kernel principle but lifts it into a learned embedding space, and it repurposes the one-class CMKKM objective as a differentiable scatter regularizer for representation learning rather than as a clustering criterion.

2.3. *Contrastive representation learning*

Contrastive learning has become a dominant paradigm for self-supervised representation learning. InfoNCE [21] and SimCLR [22] maximise agreement between augmented views of an instance against a set of negatives, MoCo [23] maintains a momentum queue of negatives, and BYOL [24] and SimSiam [25] dispense with explicit negatives altogether. Wang and Isola [26] characterise the resulting geometry through two competing properties, alignment of positive pairs and uniformity of the embedding

distribution on the hypersphere, an analysis we invoke to explain why our scatter regularizer does not collapse the representation. Whereas these methods generate views by stochastic data augmentation, we obtain views from kernel diversity, and we observe that the consistency enforced by contrastive learning does not by itself yield the intra-class compactness required for one-class detection—a gap that our scatter regularizer is designed to close.

2.4. Positioning of this work

SCMK sits at the intersection of the three threads above. From anomaly detection it inherits the one-class formulation and an OC-SVM head; from multiple kernel learning it inherits a multi-scale kernel bank and the centroid-MKKM compactness principle; and from contrastive learning it inherits a representation objective, which we adapt by treating the kernel index as the view. To our knowledge, the combination of kernel-as-view contrastive learning with a scatter regularizer that provably realises one-class CMKKM has not been explored, and it is this combination that delivers a compact, kernel-fused representation amenable to efficient linear one-class detection.

3. The proposed SCMK method

3.1. Problem formulation and notation

Let $\mathcal{X} = \{\mathbf{x}_i\}_{i=1}^N$ with $\mathbf{x}_i \in \mathbb{R}^D$ denote the data after preprocessing,¹ and let $y_i \in \{0, 1\}$ be the latent label, where $y_i = 0$ marks a normal instance and $y_i = 1$ an anomaly. Only the normal subset $\mathcal{X}_n = \{\mathbf{x}_i : y_i = 0\}$, of size $N_n = |\mathcal{X}_n|$, is available at training time; anomalies are neither observed nor simulated. The goal is to learn a scoring function $a : \mathbb{R}^D \rightarrow \mathbb{R}$ such that $a(\mathbf{x})$ is large for anomalies and small for normal points.

Our framework, termed *scatter-regularized cross-multi-kernel* contrastive learning (SCMK), couples a multiple-kernel contrastive encoder with a one-class detection head. It comprises four components, summarised in Algorithm 1: (i) a multi-scale kernel bank that injects geometric priors at several resolutions (Section 3.2); (ii)

¹Nominal attributes are one-hot encoded and numerical attributes are min–max scaled to $[0, 1]$, so that all feature dimensions share a comparable range.

a cross-kernel contrastive objective that aligns the views produced by different kernels (Section 3.3); (iii) a multi-kernel scatter regularizer that explicitly compacts the normal class in every kernel-induced embedding (Section 3.4); and (iv) a dual-signal one-class detector that fuses an angular and a radial OC-SVM score to flag anomalies (Section 3.6).

3.2. Multi-scale kernel bank

To endow the encoder with multi-resolution sensitivity, we instantiate a bank of K Gaussian kernels whose bandwidths are anchored to the intrinsic scale of the normal data. Let

$$\text{med} = \text{median}\{\|\mathbf{x}_i - \mathbf{x}_j\|_2 : \mathbf{x}_i, \mathbf{x}_j \in \mathcal{X}_n, i \neq j\} \quad (1)$$

be the median pairwise Euclidean distance estimated *from normal samples only*, which prevents contamination of the scale statistic by outliers. The k -th kernel is

$$\kappa_k(\mathbf{a}, \mathbf{b}) = \exp\left(-\frac{\|\mathbf{a} - \mathbf{b}\|_2^2}{2 t_k^2}\right), \quad t_k = r_k \cdot \text{med}, \quad (2)$$

with multiplicative ratios $\{r_k\}_{k=1}^K = \{0.1, 0.5, 1.0, 2.0, 5.0\}$ ($K = 5$). Small ratios produce sharp kernels that resolve fine-grained local neighbourhoods, whereas large ratios produce broad kernels that capture the global manifold geometry. The bank therefore provides K complementary similarity measures that the contrastive stage will reconcile.

3.3. Cross-kernel contrastive representation learning

Per-kernel embeddings. For each kernel we attach an independent, bias-free linear projection head $g_k(\mathbf{x}) = \mathbf{W}_k \mathbf{x}$ with $\mathbf{W}_k \in \mathbb{R}^{d \times D}$, followed by ℓ_2 normalization,

$$\mathbf{h}_{k,i} = \frac{\mathbf{W}_k \mathbf{x}_i}{\|\mathbf{W}_k \mathbf{x}_i\|_2} \in \mathbb{S}^{d-1}, \quad k = 1, \dots, K, \quad (3)$$

so that every embedding lives on the unit hypersphere $\mathbb{S}^{d-1}$. Removing the bias keeps the representation centred at the origin, which is consistent with the origin-separating hyperplane assumption of the downstream OC-SVM, while the normalization equalises the scale across kernels.

Cross-kernel positive pairs. Whereas conventional contrastive learning forms positive pairs through stochastic data augmentation, we treat the *kernel index as the view*: the two embeddings $\mathbf{h}_{k,i}$ and $\mathbf{h}_{l,i}$ of the *same* sample under *different* kernels constitute a positive pair, and embeddings of distinct samples are negatives. Consider a mini-batch of B normal samples and a kernel pair (k, l) . Stacking the two views yields $\mathbf{F}^{(kl)} = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,B}, \mathbf{h}_{l,1}, \dots, \mathbf{h}_{l,B}]^\top \in \mathbb{R}^{2B \times d}$, whose rows are indexed by $m, n \in \{1, \dots, 2B\}$. The pairwise affinity is the *symmetrised kernel response* evaluated on the embeddings,

$$s_{mn}^{(kl)} = \frac{1}{2} [\kappa_k(\mathbf{f}_m, \mathbf{f}_n) + \kappa_l(\mathbf{f}_m, \mathbf{f}_n)]. \quad (4)$$

Because the embeddings are unit-norm, the Gaussian response reduces to a monotone function of cosine similarity, $\kappa_k(\mathbf{f}_m, \mathbf{f}_n) = \exp(-(1 - \langle \mathbf{f}_m, \mathbf{f}_n \rangle)/t_k^2)$, so t_k^2 plays the role of a contrastive temperature that is automatically adapted to the data scale through Eq. (1).

Cross-kernel InfoNCE. For anchor n , let $\pi(n)$ index its cross-view positive (i.e. $\pi(n) = n + B$ for $n \leq B$ and $\pi(n) = n - B$ otherwise). The InfoNCE loss for the pair (k, l) is

$$\mathcal{L}_{\text{con}}^{(kl)} = -\frac{1}{2B} \sum_{n=1}^{2B} \log \frac{\exp(s_{n,\pi(n)}^{(kl)})}{\sum_{m \neq n} \exp(s_{nm}^{(kl)})}, \quad (5)$$

where the denominator excludes the self-similarity term $m = n$. Averaging over all $\binom{K}{2}$ unordered kernel pairs gives the cross-kernel contrastive objective

$$\mathcal{L}_{\text{con}} = \frac{1}{\binom{K}{2}} \sum_{1 \leq k < l \leq K} \mathcal{L}_{\text{con}}^{(kl)}. \quad (6)$$

Minimising Eq. (6) drives the encoder to a *view-invariant* representation in which all kernels agree on the embedding of each normal sample, i.e. $\mathbf{h}_{k,i} \approx \mathbf{h}_{l,i}$, while keeping distinct samples mutually discriminable through the negative terms.

3.4. Multi-kernel scatter regularization

Motivation. The contrastive objective (6) enforces *cross-kernel consistency* but does not constrain the *absolute compactness* of the normal class: it is invariant to a rotation that spreads normal embeddings across the sphere as long as the per-sample

views remain aligned. A one-class detector, however, benefits directly from a tightly concentrated normal region. We therefore introduce a regularizer that minimises the dispersion of the normal class within every kernel-induced embedding space, drawing on the one-class specialisation of centroid multiple-kernel k -means (CMKKM).

Definition.. Let $\boldsymbol{\mu}_k = \frac{1}{B} \sum_{i=1}^B \mathbf{h}_{k,i}$ be the empirical centroid of the normal embeddings under kernel k . The multi-kernel scatter penalty is

$$\mathcal{L}_{\text{scat}} = -\frac{1}{K} \sum_{k=1}^K \|\boldsymbol{\mu}_k\|_2^2. \quad (7)$$

It is computed in $O(KBd)$ time — a single mean per kernel — and thus adds negligible overhead.

Exact equivalence to within-class scatter.. The next result shows that Eq. (7) is not a heuristic surrogate but is *exactly* (up to an additive constant) the average within-class scatter of the normal embeddings.

Proposition 1 (Scatter equivalence). *For unit-norm embeddings $\mathbf{h}_{k,i} \in \mathbb{S}^{d-1}$, define the average normalised within-class scatter $\sigma^2 = \frac{1}{NK} \sum_{k=1}^K \sum_{i=1}^N \|\mathbf{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2$. Then*

$$\sigma^2 = 1 + \mathcal{L}_{\text{scat}}. \quad (8)$$

Consequently, minimising $\mathcal{L}_{\text{scat}}$ is equivalent to minimising the within-class scatter σ^2 , and equivalently to maximising the mean intra-class kernel alignment $\frac{1}{NK} \sum_k \sum_{i,j} \langle \mathbf{h}_{k,i}, \mathbf{h}_{k,j} \rangle = \frac{1}{K} \sum_k \|\boldsymbol{\mu}_k\|_2^2$.

Proof. Fix a kernel k . Expanding the squared norm and using $\|\mathbf{h}_{k,i}\|_2^2 = 1$,

$$\begin{aligned} \sum_{i=1}^N \|\mathbf{h}_{k,i} - \boldsymbol{\mu}_k\|_2^2 &= \sum_{i=1}^N (\|\mathbf{h}_{k,i}\|_2^2 - 2 \langle \mathbf{h}_{k,i}, \boldsymbol{\mu}_k \rangle + \|\boldsymbol{\mu}_k\|_2^2) \\ &= N - 2N \|\boldsymbol{\mu}_k\|_2^2 + N \|\boldsymbol{\mu}_k\|_2^2 = N(1 - \|\boldsymbol{\mu}_k\|_2^2), \end{aligned} \quad (9)$$

where $\sum_i \langle \mathbf{h}_{k,i}, \boldsymbol{\mu}_k \rangle = N \langle \boldsymbol{\mu}_k, \boldsymbol{\mu}_k \rangle = N \|\boldsymbol{\mu}_k\|_2^2$ follows from the definition of $\boldsymbol{\mu}_k$. Averaging Eq. (9) over k and dividing by N yields $\sigma^2 = 1 - \frac{1}{K} \sum_k \|\boldsymbol{\mu}_k\|_2^2 = 1 + \mathcal{L}_{\text{scat}}$. The alignment identity is immediate from $\sum_{i,j} \langle \mathbf{h}_{k,i}, \mathbf{h}_{k,j} \rangle = N^2 \|\boldsymbol{\mu}_k\|_2^2$. $\square$

Remark 1 (Connection to one-class CMKKM). *The one-class instance of centroid multiple-kernel k -means maximises $\text{Tr}(\mathbf{H}^\top \mathbf{K}_\eta \mathbf{H})$ with a single soft cluster indicator $\mathbf{H} = N^{-1/2} \mathbf{1}$ and a convex kernel combination $\mathbf{K}_\eta = \sum_k \eta_k \Phi_k \Phi_k^\top$, where $\Phi_k = [\mathbf{h}_{k,1}, \dots, \mathbf{h}_{k,N}]^\top$ realises a linear kernel on the embeddings. A short calculation gives $\text{Tr}(\mathbf{H}^\top \mathbf{K}_\eta \mathbf{H}) = N \sum_k \eta_k \|\boldsymbol{\mu}_k\|_2^2$, which under uniform weights $\eta_k = 1/K$ equals $-N \mathcal{L}_{\text{scat}}$. Hence $\mathcal{L}_{\text{scat}}$ is precisely the (negated, scale-normalised) one-class CMKKM compactness objective, providing a multiple-kernel-learning justification for the penalty.*

3.5. Joint objective and collapse avoidance

The encoder $\{\mathbf{W}_k\}_{k=1}^K$ is trained by minimising the combined loss

$$\mathcal{L} = \mathcal{L}_{\text{con}} + \lambda \mathcal{L}_{\text{scat}}, \quad (10)$$

where $\lambda \geq 0$ trades off cross-kernel consistency against intra-class compactness; $\lambda = 0$ recovers the purely contrastive backbone.

A natural concern is representational *collapse*: the scatter term alone is trivially minimised ($\mathcal{L}_{\text{scat}} = -1$) when all embeddings of a kernel coincide ($\|\boldsymbol{\mu}_k\|_2 = 1$). This degenerate solution is, however, suppressed by the contrastive term. In the alignment–uniformity decomposition of contrastive learning, the numerator of Eq. (5) promotes *alignment* of positives while the log-sum-exp denominator promotes *uniformity* of the embedding distribution on the sphere, which is maximised only by a spread-out configuration and is therefore directly opposed to collapse. The joint objective (10) thus realises a controlled balance: $\mathcal{L}_{\text{scat}}$ pulls each normal class towards its centroid, $\mathcal{L}_{\text{con}}$ prevents the normal embeddings from degenerating to a point and keeps different samples separable, and λ governs the equilibrium. The unit-sphere constraint additionally bounds $\|\boldsymbol{\mu}_k\|_2 \leq 1$, so the penalty cannot diverge.

3.6. Dual-signal anomaly scoring

After training, the projection heads are frozen and each sample $\mathbf{x}_i$ is described by *two complementary signals* read off the raw projections $\mathbf{W}_k \mathbf{x}_i$: the *angular* position of its unit-norm embeddings and the *radial* magnitude of those projections. These signals carry non-redundant discriminative information, and which of them separates

the normal and anomalous classes is dataset-dependent. We therefore build a detector for each and fuse their scores.

Directional signal: linear OC-SVM on the angular embedding. The unit-norm embeddings $\mathbf{h}_{k,i}$ retain only the angular position of a sample on each kernel sphere—exactly the geometry that the scatter regularizer (7) compacts. Concatenating the K views,

$$\mathbf{z}_i = [\mathbf{h}_{1,i}^\top, \mathbf{h}_{2,i}^\top, \dots, \mathbf{h}_{K,i}^\top]^\top \in \mathbb{R}^{Kd}, \quad (11)$$

fuses them into a single descriptor, and a linear-kernel OC-SVM fitted on the normal embeddings $\{\mathbf{z}_i : y_i = 0\}$ by solving

$$\min_{\mathbf{w}, \rho, \xi} \frac{1}{2} \|\mathbf{w}\|_2^2 - \rho + \frac{1}{vN_n} \sum_{i: y_i=0} \xi_i \quad \text{s.t.} \quad \langle \mathbf{w}, \mathbf{z}_i \rangle \geq \rho - \xi_i, \quad \xi_i \geq 0, \quad (12)$$

yields the *directional score* $s_i^{\text{dir}} = \rho - \langle \mathbf{w}, \mathbf{z}_i \rangle$, where $v \in (0, 1]$ upper-bounds the fraction of normal training points allowed outside the decision region. A linear separator suffices here because the contrastive and scatter objectives jointly mould the normal class into a compact, nearly linearly separable cone on the product of spheres $(\mathbb{S}^{d-1})^K$; this score is informative whenever the normal/anomaly contrast is *angular*.

Magnitude signal: RBF OC-SVM on the projection norms. The ℓ_2 normalization in Eq. (3) deliberately discards the projection norms $r_{k,i} = \|\mathbf{W}_k \mathbf{x}_i\|_2$. Yet these norms are themselves discriminative: trained on normal data alone, each head maps normal samples into a characteristic energy band, whereas anomalies—lying off the learned manifold—are projected to markedly different, typically larger, norms. We collect the per-kernel norms into a magnitude vector

$$\mathbf{r}_i = [r_{1,i}, r_{2,i}, \dots, r_{K,i}]^\top \in \mathbb{R}^K, \quad (13)$$

and fit an *RBF*-kernel OC-SVM on the normal norms $\{\mathbf{r}_i : y_i = 0\}$, producing the *magnitude score* s_i^{mag} . The RBF kernel is essential: a *linear* OC-SVM on norms admits a sign-reversed optimum—when the normal norms are small, the separating direction turns negative and labels large-norm anomalies as inliers—whereas the RBF density model, centred on the compact small-norm normal cluster, correctly assigns high scores

to the large-norm outliers lying outside it. This score is informative whenever the contrast is *radial*.

Score fusion.. On a given dataset one signal may be essentially uninformative—most strikingly, the directional score collapses to near-chance when normal and anomalous samples are angularly indistinguishable but differ in projection energy. Without assuming *a priori* which regime holds, we fuse the two detectors by a per-sample maximum of min–max-normalised scores,

$$a(\mathbf{x}_i) = \max(\tilde{s}_i^{\text{dir}}, \tilde{s}_i^{\text{mag}}), \quad \tilde{s}_i^\bullet = \frac{s_i^\bullet - \min_j s_j^\bullet}{\max_j s_j^\bullet - \min_j s_j^\bullet}, \quad (14)$$

so that a sample flagged by *either* detector receives a high anomaly score. The fusion needs no prior knowledge of which signal is informative and, as the ablation in Section 4.6 confirms, remains robust on datasets where either single signal collapses while never trailing the stronger signal by a meaningful margin. Fig. 1 makes this complementarity geometric: on a direction-dominant dataset the anomalies stay on the sphere but cluster to one angular region, whereas on a magnitude-dominant dataset they are angularly mixed with the normal class yet bulge far outside the sphere—so a single detector necessarily fails on one regime, while the max-fusion captures both.

3.7. Overall algorithm and complexity

Algorithm 1 summarises the complete training and scoring pipeline. The dominant per-mini-batch cost is the cross-kernel contrastive term, $\mathcal{O}(K^2 B^2 d)$ for the $\binom{K}{2}$ pairwise affinity matrices, while the scatter regularizer contributes only $\mathcal{O}(K B d)$. Embedding extraction over the full data set is $\mathcal{O}(N K d D)$. Scoring fits two OC-SVMs on the normal set: a linear one on the Kd -dimensional embeddings and an RBF one on the K -dimensional magnitude vectors; both scale with the number of support vectors, and the radial detector is negligible since $K = 5$. With a fixed small kernel bank, the method retains the same asymptotic complexity as the contrastive backbone, so the accuracy gains reported in the experiments come at no additional order-of-growth cost.

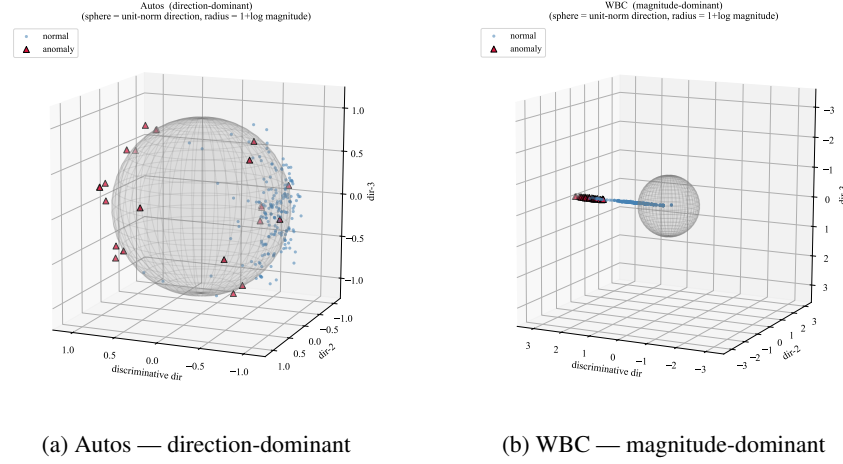


Fig. 1. Geometric view of the two complementary signals on the learned representation. The grey sphere is the unit-norm (direction) manifold; a sample’s radius is $1 + \log$ of its projection magnitude, so normal points (≈ 1) lie on the shell. **(a)** On Autos the anomalies remain on the shell but concentrate in one angular region—separable by *direction*, while their magnitude is uninformative. **(b)** On WBC the anomalies are angularly intermixed with the normal class yet bulge far outside the shell—separable only by *magnitude*. The angular (linear OC-SVM) detector handles (a), the radial (RBF OC-SVM) detector handles (b), and the max-fusion is robust on both without knowing the regime in advance. For visualization only, the principal in-sphere axis is the normal/anomaly mean-difference direction (labels are used solely for this projection, never for training or scoring).

4. Experiments

4.1. Experimental setup

We evaluate on twenty public benchmark datasets² drawn from the UCI repository, summarised in Table 1. They span a wide range of sample sizes (101–5171), dimensionalities (6–154), anomaly rates (1.9%–25.2%), scenarios, and cover numerical, nominal and mixed-type attributes, so that robustness can be assessed across heterogeneous regimes. Following the protocol adopted in recent anomaly-detection studies, each dataset is split so that 50% of the normal samples form the training set while the remaining 50% of normal samples together with all anomalies form the test set. For the one-class methods—SCMK, Disent-AD, DeepSVDD, LMKAD, ICL and

²<https://github.com/BELLoney/Outlier-detection>

Algorithm 1 SCMK for one-class anomaly detection

Require: normal set $\mathcal{X}_n$, full set $\mathcal{X}$, kernel ratios $\{r_k\}_{k=1}^K$, embedding dim. d , weight λ , batch size B , epochs T , OC-SVM parameter ν

Ensure: anomaly scores $\{a(\mathbf{x}_i)\}_{i=1}^N$

- 1: $\text{med} \leftarrow$ median pairwise distance on $\mathcal{X}_n$ ▷ Eq. (1)
 - 2: build kernels κ_k with $t_k = r_k \text{ med}$ ▷ Eq. (2)
 - 3: initialise projection heads $\{\mathbf{W}_k\}_{k=1}^K$
 - 4: **for** epoch = 1 **to** T **do**
 - 5: **for** each mini-batch of B samples from $\mathcal{X}_n$ **do**
 - 6: $\mathbf{h}_{k,i} \leftarrow \mathbf{W}_k \mathbf{x}_i / \|\mathbf{W}_k \mathbf{x}_i\|_2$ for all k, i ▷ Eq. (3)
 - 7: $\mathcal{L}_{\text{con}} \leftarrow$ cross-kernel InfoNCE over kernel pairs ▷ Eqs. (4)–(6)
 - 8: $\mathcal{L}_{\text{scat}} \leftarrow -\frac{1}{K} \sum_k \left\| \frac{1}{B} \sum_i \mathbf{h}_{k,i} \right\|_2^2$ ▷ Eq. (7)
 - 9: update $\{\mathbf{W}_k\}$ by a gradient step on $\mathcal{L}_{\text{con}} + \lambda \mathcal{L}_{\text{scat}}$ ▷ Eq. (10)
 - 10: **end for**
 - 11: **end for**
 - 12: $\mathbf{z}_i \leftarrow [\mathbf{h}_{1,i}; \dots; \mathbf{h}_{K,i}]$, $\mathbf{r}_i \leftarrow [\|\mathbf{W}_1 \mathbf{x}_i\|; \dots; \|\mathbf{W}_K \mathbf{x}_i\|]$ for all $\mathbf{x}_i \in \mathcal{X}$ ▷ Eqs. (11),(13)
 - 13: $s_i^{\text{dir}} \leftarrow$ linear OC-SVM on $\{\mathbf{z}_i : y_i = 0\}$; $s_i^{\text{mag}} \leftarrow$ RBF OC-SVM on $\{\mathbf{r}_i : y_i = 0\}$ ▷
Eq. (12)
 - 14: **return** $a(\mathbf{x}_i) = \max(\tilde{s}_i^{\text{dir}}, \tilde{s}_i^{\text{mag}})$ ▷ Eq. (14)
-

NeuTraLAD—we repeat it for three random seeds (0, 1, 2), training on the training set and scoring on the test set, and report the mean $_{\pm\text{std}}$ AUC; KFGOD and DFNO score samples transductively without a training phase and run on the full set as in their native protocol.

We compare against seven recent outlier detectors: KFGOD [31], Disent-AD [30], DeepSVDD [29], DFNO [28], LMKAD [32], ICL [33] and NeuTraLAD [34]. These include fuzzy-rough/neighbourhood (KFGOD, DFNO), deep one-class representation (Disent-AD, DeepSVDD), multiple-kernel (LMKAD), self-supervised contrastive and transformation (ICL, NeuTraLAD) paradigms, providing a strong and diverse modern reference set.

Table 1. The twenty benchmark datasets (N : samples, D : features, ρ : anomaly rate).

Dataset	N	D	ρ	Dataset	N	D	ρ
Vertebral	240	6	12.5%	TicTacToe-69	695	27	9.9%
Thyroid	3772	6	2.5%	WPBC	198	33	23.7%
Ecoli	336	9	2.7%	Ionosphere	249	35	9.6%
Glass	214	9	4.2%	Zoo	101	36	16.8%
WBC	483	9	8.1%	Sick-72	3613	53	2.0%
PageBlocks	5171	10	5.0%	Autos	205	54	12.2%
Wine	129	13	7.8%	Annealing	798	56	5.3%
Cardio	1831	21	9.6%	Lympho.	148	59	4.0%
Cardiotoco.	1688	22	1.9%	Bands-6	318	62	1.9%
TicTacToe-12	638	27	1.9%	Audiology	226	154	25.2%

The encoder of SCMK uses 5 Gaussian kernels with ratios $\{0.1, 0.5, 1, 2, 5\}$ and bias-free linear heads, trained for 100 epochs with Adam (learning rate 0.01, batch size 512) on normal data only; the random seed is fixed to 42. The embedding dimension $d \in \{16, 32, 64, 128, 256\}$, the scatter weight $\lambda \in \{0, 0.1, 1, 10, 100, 1000\}$ and the OC-SVM parameter $\nu \in \{0.01, 0.05, 0.1, 0.2\}$ are selected per dataset. As is customary on these benchmarks, every method—ours and all baselines—is reported at its best hyper-parameter configuration, so the comparison reflects each detector’s attainable ceiling under a common protocol.

Detection quality is measured by the area under the ROC curve (AUC), with normal samples labelled 0 and anomalies 1; higher is better.

4.2. Comparison with recent detectors

Table 2 reports the AUC of SCMK and the seven baselines. SCMK attains the highest mean AUC, $0.914_{\pm 0.022}$, exceeding the strongest single competitor (DFNO, 0.863) by 5.1 points, the next one-class methods LMKAD (0.850) and NeuTraLAD (0.835) by 6.4 and 7.9 points, and the seven-method average (0.824) by 9.0 points; it also has the lowest across-seed standard deviation (0.022) among the one-class methods. SCMK holds the best mean on 9 of the 20 datasets and wins 114 of the 140 pair-

wise comparisons. The advantage is clearest where the baselines collectively struggle: on Cardiotocography the best competitor reaches only 0.873 against SCMK’s 0.914, on Annealing 0.843 against 0.877, and on Audiology 0.970 against 0.982. On the eleven datasets where SCMK does not hold the best mean it is usually a close second—within 0.01–0.02 of the leader (e.g. Thyroid 0.981 vs. LMKAD/Disent-AD 0.985, Cardio 0.969 vs. LMKAD 0.973, TicTacToe-12 0.990 vs. DFNO 1.000, TicTacToe-69 0.981 vs. DFNO 0.999); the only sizeable shortfalls are Glass and Vertebral, where Disent-AD’s deep one-class encoder reaches 0.936 and 0.785. WBC is an angularly indistinguishable case—used to motivate the magnitude branch in Fig. 1—on which SCMK still attains 0.992. That SCMK leads on average across numerical, nominal and mixed-type data, with the lowest across-seed standard deviation (0.022) among the one-class methods, indicates that the kernel-fused, scatter-compacted representation generalises robustly across feature types and random splits rather than exploiting a single data regime. Per-dataset ROC curves for all twenty datasets are given in Fig. 2. The statistical significance of this advantage is established in Section 4.4. As a threshold-based complement to the ranking-based AUC, Table 3 reports the maximum G-mean ($\sqrt{\text{TPR} \cdot \text{TNR}}$) of every method; SCMK again attains the highest mean (0.863) and the most first-place results, confirming that its advantage persists once a concrete decision threshold is imposed and is not an artefact of the threshold-free AUC.

4.3. Runtime evaluation

Because SCMK forms $\binom{K}{2}$ pairwise cross-kernel affinity matrices per mini-batch and maintains K projection heads, its cost warrants an empirical check rather than an asymptotic argument alone. We therefore measure the actual wall-clock *training* and *inference* time of SCMK against the baselines under a unified protocol: the same GPU, the seed-2 split, a single per-dataset configuration, one CUDA warm-up, and the median of repeated inference passes. Training is the fit on the normal training set; inference is scoring the test set. The transductive detectors (KFGOD, DFNO) have no separate training phase, and to isolate the multiple-kernel overhead we also time the single-kernel variant SCMK_{K=1}. KFGOD and LMKAD are timed through their official code; the results are summarised in Fig. 3 (means over the twenty datasets).

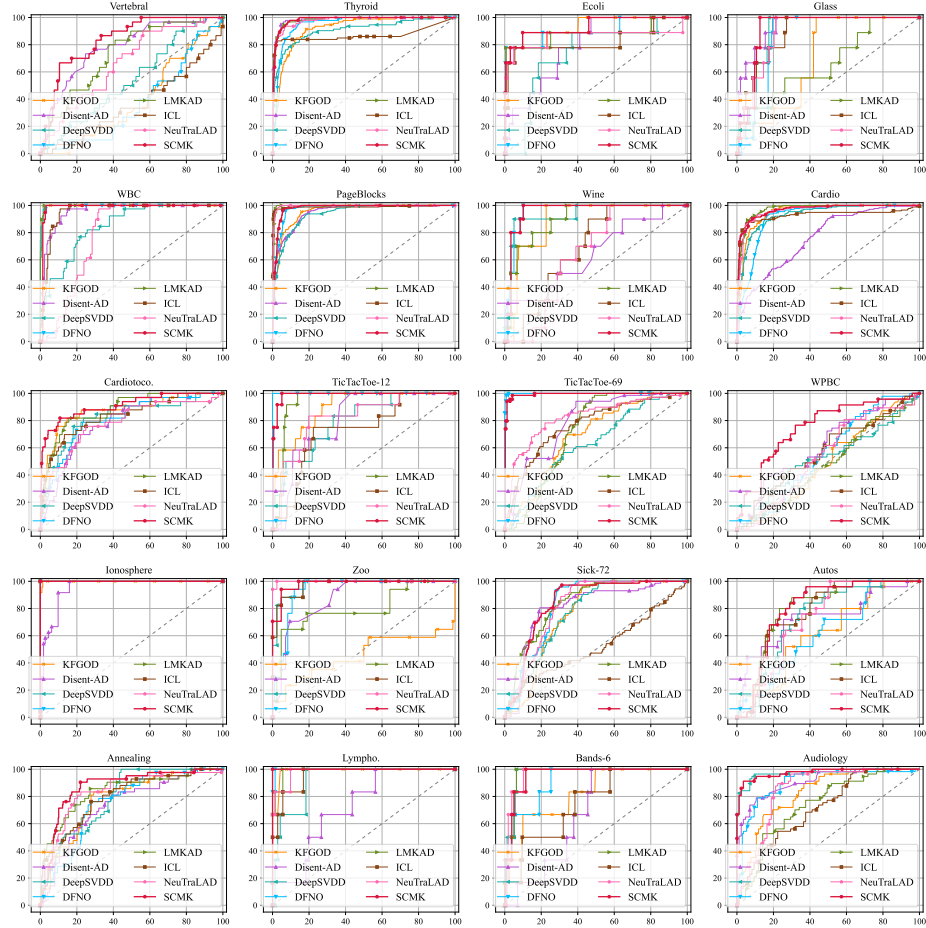


Fig. 2. Per-dataset ROC curves on the twenty datasets.

Table 2. AUC of SCMCK against seven recent detectors on twenty datasets. The one-class methods (SCMK, Disent-AD, DeepSVDD, LMKAD, ICL, NeuTraLAD) are reported as mean $\pm$ std over three random splits (seeds 0, 1, 2). Best mean in each row is in **bold**.

Dataset	KFGOD [31]	Disent [30]	DeepSVDD [29]	DFNO [28]	LMKAD [32]	ICL [33]	NeuTraLAD [34]	SCMK (ours)
Vertebral	0.390	0.785 ± 0.026	0.517 ± 0.024	0.373	0.655 ± 0.067	0.345 ± 0.060	0.627 ± 0.043	0.735 ± 0.092
Thyroid	0.921	0.985 ± 0.001	0.918 ± 0.010	0.953	0.985 ± 0.002	0.880 ± 0.035	0.979 ± 0.001	0.981 ± 0.003
WBC	0.997	0.964 ± 0.004	0.864 ± 0.024	0.997	0.992 ± 0.003	0.967 ± 0.008	0.808 ± 0.034	0.992 ± 0.007
Glass	0.700	0.936 ± 0.002	0.875 ± 0.018	0.857	0.588 ± 0.069	0.855 ± 0.038	0.875 ± 0.012	0.853 ± 0.058
Ecoli	0.926	0.834 ± 0.041	0.761 ± 0.042	0.905	0.814 ± 0.110	0.832 ± 0.006	0.866 ± 0.011	0.931 ± 0.010
PageBlocks	0.957	0.937 ± 0.010	0.945 ± 0.019	0.975	0.995 ± 0.001	0.989 ± 0.004	0.996 ± 0.001	0.976 ± 0.004
Wine	0.887	0.633 ± 0.026	0.948 ± 0.015	0.957	0.905 ± 0.011	0.794 ± 0.122	0.639 ± 0.025	0.920 ± 0.037
Cardio	0.950	0.744 ± 0.018	0.935 ± 0.016	0.914	0.973 ± 0.003	0.896 ± 0.029	0.960 ± 0.003	0.969 ± 0.002
Cardiotoco.	0.870	0.804 ± 0.013	0.839 ± 0.013	0.809	0.873 ± 0.014	0.751 ± 0.102	0.754 ± 0.024	0.914 ± 0.008
TicTacToe-12	0.898	0.811 ± 0.111	0.750 ± 0.043	1.000	0.938 ± 0.004	0.694 ± 0.067	0.823 ± 0.009	0.990 ± 0.003
TicTacToe-69	0.690	0.766 ± 0.020	0.626 ± 0.028	0.999	0.707 ± 0.037	0.739 ± 0.042	0.815 ± 0.010	0.981 ± 0.018
WPBC	0.494	0.631 ± 0.008	0.532 ± 0.004	0.556	0.470 ± 0.016	0.564 ± 0.039	0.478 ± 0.070	0.653 ± 0.090
Ionosphere	0.999	0.954 ± 0.017	1.000 ± 0.001	1.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
Zoo	0.457	0.883 ± 0.009	0.939 ± 0.064	0.934	0.842 ± 0.047	0.959 ± 0.040	0.968 ± 0.050	0.969 ± 0.020
Sick-72	0.760	0.815 ± 0.009	0.774 ± 0.038	0.833	0.846 ± 0.049	0.582 ± 0.174	0.818 ± 0.003	0.828 ± 0.016
Autos	0.625	0.722 ± 0.035	0.679 ± 0.048	0.623	0.804 ± 0.021	0.725 ± 0.083	0.707 ± 0.049	0.822 ± 0.019
Annealing	0.783	0.735 ± 0.009	0.795 ± 0.044	0.759	0.842 ± 0.018	0.758 ± 0.062	0.843 ± 0.016	0.877 ± 0.008
Lympho.	0.988	0.745 ± 0.017	0.931 ± 0.037	0.998	0.995 ± 0.006	0.947 ± 0.021	0.976 ± 0.007	0.999 ± 0.001
Bands-6	0.822	0.677 ± 0.012	0.936 ± 0.047	0.909	0.941 ± 0.025	0.789 ± 0.033	0.953 ± 0.021	0.914 ± 0.044
Audiology	0.831	0.887 ± 0.020	0.970 ± 0.009	0.903	0.831 ± 0.079	0.681 ± 0.039	0.819 ± 0.074	0.982 ± 0.008
Average	0.797	0.812 ± 0.020	0.827 ± 0.027	0.863	0.850 ± 0.029	0.787 ± 0.050	0.835 ± 0.023	0.914 ± 0.022

The multiple-kernel overhead is real but bounded. SCMCK trains in 3.07 s on average versus 0.23 s for SCMCK $_{K=1}$, so the $\binom{K}{2}$ affinity computation with $K = 5$ multiplies training time by roughly 13 $\times$. In absolute terms, however, SCMCK remains firmly mid-pack among the deep detectors: it is faster than NeuTraLAD (4.80 s) and about 5 $\times$ faster than ICL (15.25 s), and comparable to Disent-AD (2.66 s), while the lighter DeepSVDD (1.04 s) and the kernel one-class LMKAD (0.35 s) are cheaper. SCMCK is thus not an outlier in training cost, and the overhead buys the multi-scale representation that drives its accuracy.

Inference is near-instant and scales gracefully. At test time SCMCK scores a dataset in 11 ms on average—among the fastest methods, since inference is a single forward

Table 3. Maximum G-mean ($\sqrt{\text{TPR} \cdot \text{TNR}}$) of SCMK against seven recent detectors on twenty datasets, as a threshold-based complement to the AUC in Table 2. The one-class methods (SCMK, Disent-AD, DeepSVDD, LMKAD, ICL, NeuTraLAD) are reported as mean $\pm$ std over three random splits (seeds 0, 1, 2); KFGOD and DFNO are transductive on the full set (single-valued). Best mean in each row is in **bold**.

Dataset	KFGOD	Disent	DeepSVDD	DFNO	LMKAD	ICL	NeuTraLAD	SCMK
Vertebral	0.454	0.741 $\pm$ 0.022	0.533 $\pm$ 0.005	0.436	0.661 $\pm$ 0.047	0.423 $\pm$ 0.055	0.624 $\pm$ 0.052	0.669 $\pm$ 0.106
Thyroid	0.864	0.952 $\pm$ 0.003	0.855 $\pm$ 0.030	0.909	0.947 $\pm$ 0.005	0.882 $\pm$ 0.051	0.935 $\pm$ 0.002	0.936 $\pm$ 0.007
WBC	0.992	0.926 $\pm$ 0.015	0.797 $\pm$ 0.025	0.985	0.986 $\pm$ 0.004	0.937 $\pm$ 0.004	0.816 $\pm$ 0.030	0.985 $\pm$ 0.003
Glass	0.752	0.892 $\pm$ 0.025	0.896 $\pm$ 0.011	0.897	0.613 $\pm$ 0.024	0.843 $\pm$ 0.044	0.889 $\pm$ 0.003	0.861 $\pm$ 0.073
Ecoli	0.878	0.798 $\pm$ 0.054	0.781 $\pm$ 0.059	0.879	0.821 $\pm$ 0.101	0.866 $\pm$ 0.008	0.866 $\pm$ 0.018	0.890 $\pm$ 0.024
PageBlocks	0.895	0.872 $\pm$ 0.009	0.874 $\pm$ 0.031	0.948	0.987 $\pm$ 0.002	0.974 $\pm$ 0.005	0.979 $\pm$ 0.001	0.953 $\pm$ 0.003
Wine	0.879	0.641 $\pm$ 0.046	0.935 $\pm$ 0.012	0.948	0.857 $\pm$ 0.035	0.797 $\pm$ 0.107	0.673 $\pm$ 0.018	0.720 $\pm$ 0.227
Cardio	0.890	0.681 $\pm$ 0.009	0.865 $\pm$ 0.017	0.879	0.924 $\pm$ 0.004	0.852 $\pm$ 0.052	0.903 $\pm$ 0.008	0.910 $\pm$ 0.010
Cardiotoco.	0.835	0.781 $\pm$ 0.036	0.793 $\pm$ 0.006	0.778	0.820 $\pm$ 0.009	0.716 $\pm$ 0.082	0.719 $\pm$ 0.030	0.836 $\pm$ 0.022
TicTacToe-12	0.821	0.795 $\pm$ 0.093	0.714 $\pm$ 0.030	1.000	0.933 $\pm$ 0.004	0.710 $\pm$ 0.072	0.811 $\pm$ 0.013	0.747 $\pm$ 0.267
TicTacToe-69	0.677	0.751 $\pm$ 0.011	0.605 $\pm$ 0.020	0.990	0.714 $\pm$ 0.027	0.689 $\pm$ 0.028	0.774 $\pm$ 0.015	0.965 $\pm$ 0.034
WPBC	0.539	0.629 $\pm$ 0.022	0.554 $\pm$ 0.012	0.561	0.505 $\pm$ 0.009	0.594 $\pm$ 0.051	0.517 $\pm$ 0.067	0.634 $\pm$ 0.076
Ionosphere	0.993	0.903 $\pm$ 0.031	0.997 $\pm$ 0.005	1.000	1.000 $\pm$ 0.000	0.997 $\pm$ 0.005	1.000 $\pm$ 0.000	0.996 $\pm$ 0.008
Zoo	0.529	0.843 $\pm$ 0.051	0.935 $\pm$ 0.047	0.911	0.802 $\pm$ 0.038	0.922 $\pm$ 0.039	0.963 $\pm$ 0.033	0.913 $\pm$ 0.048
Sick-72	0.744	0.811 $\pm$ 0.005	0.726 $\pm$ 0.043	0.821	0.782 $\pm$ 0.049	0.584 $\pm$ 0.171	0.799 $\pm$ 0.015	0.816 $\pm$ 0.010
Autos	0.624	0.716 $\pm$ 0.014	0.684 $\pm$ 0.046	0.613	0.806 $\pm$ 0.021	0.720 $\pm$ 0.073	0.692 $\pm$ 0.059	0.780 $\pm$ 0.021
Annealing	0.768	0.697 $\pm$ 0.012	0.751 $\pm$ 0.025	0.743	0.800 $\pm$ 0.035	0.707 $\pm$ 0.049	0.792 $\pm$ 0.013	0.812 $\pm$ 0.025
Lympho.	0.979	0.717 $\pm$ 0.018	0.919 $\pm$ 0.033	0.993	0.988 $\pm$ 0.015	0.915 $\pm$ 0.011	0.944 $\pm$ 0.004	0.995 $\pm$ 0.004
Bands-6	0.775	0.692 $\pm$ 0.030	0.927 $\pm$ 0.073	0.864	0.940 $\pm$ 0.029	0.763 $\pm$ 0.043	0.932 $\pm$ 0.038	0.896 $\pm$ 0.045
Audiology	0.754	0.825 $\pm$ 0.017	0.933 $\pm$ 0.021	0.837	0.784 $\pm$ 0.080	0.653 $\pm$ 0.016	0.806 $\pm$ 0.066	0.948 $\pm$ 0.022
Average	0.782	0.783 $\pm$ 0.026	0.804 $\pm$ 0.028	0.850	0.834 $\pm$ 0.027	0.777 $\pm$ 0.048	0.822 $\pm$ 0.024	0.863 $\pm$ 0.052

pass plus two linear/RBF OC-SVM evaluations. The two transductive baselines are orders of magnitude slower and, being $O(n^2)$, degrade sharply with size: on the largest test set (PageBlocks, $n = 2715$) SCMK infers in 0.13 s whereas DFNO takes 2.47 s and KFGOD 27.97 s. SCMK therefore pays a modest one-off training premium for its kernel bank yet is highly efficient to deploy, in contrast to transductive methods that must recompute over the whole data at every query.

4.4. Statistical significance

To confirm that the advantage in Table 2 is statistically significant rather than incidental, we follow the standard protocol of Demšar [27]: a Friedman test across all eight methods (using each method’s per-dataset mean AUC), followed by the Nemenyi

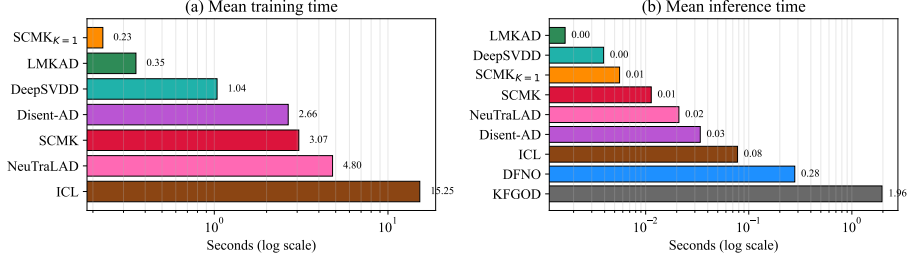


Fig. 3. Mean (a) training and (b) inference time (seconds, log scale) over the twenty datasets on a single GPU. SCM_{K=1} is the single-kernel variant; KFGOD and DFNO are transductive and appear only under inference. SCM’s multiple-kernel training cost (3.07 s) stays below NeuTraLAD and ICL, and its inference (11 ms) is among the fastest—far below the transductive KFGOD (1.96 s) and DFNO (0.28 s).

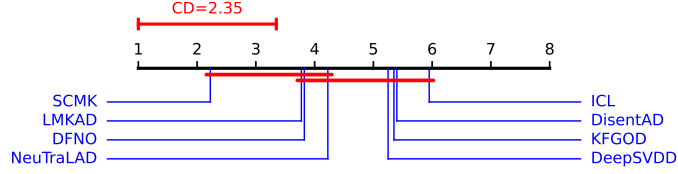


Fig. 4. Critical-difference diagram (Nemenyi post-hoc, $\alpha = 0.05$, $CD = 2.35$) over the twenty datasets, using each method’s per-dataset mean AUC. Lower rank (towards the right) is better; methods not connected by a thick bar are not significantly different. SCM (mean rank 2.23) has the best rank; DisentAD, DeepSVDD, KFGOD and ICL are significantly worse, while LMKAD, DFNO and NeuTraLAD share its top cluster.

post-hoc test rendered as a critical-difference (CD) diagram.

The Friedman test strongly rejects the null hypothesis that the eight methods perform equally ($\chi^2_F = 34.98$, $p = 1.1 \times 10^{-5}$; Iman–Davenport $F = 6.33$). Fig. 4 shows the CD diagram at significance level $\alpha = 0.05$ ($CD = 2.35$). SCM attains the best mean rank, 2.23, and is separated by more than the critical difference from four of the seven baselines (DeepSVDD 5.25, KFGOD 5.35, DisentAD 5.40, ICL 5.95). The three strongest competitors—LMKAD (3.78), DFNO (3.83) and NeuTraLAD (4.23)—fall within the critical difference of SCM, so the conservative Nemenyi test places them in the same top cluster; the pairwise analysis below resolves this finer ordering and still finds SCM significantly ahead of all three.

We further corroborate this with pairwise one-sided Wilcoxon signed-rank tests of

Table 4. Pairwise comparison of SCMK against each baseline (ordered by average rank): mean rank, number of datasets on which SCMK wins, and one-sided Wilcoxon signed-rank p -values before and after Holm correction. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$.

Method	Avg. rank	SCMK wins	Wilcoxon p	Holm p
LMKAD [32]	3.78	14/20	4.5×10^{-3}	8.9×10^{-3} **
DFNO [28]	3.83	13/20	1.7×10^{-2}	1.7×10^{-2} *
NeuTraLAD [34]	4.23	16/20	1.3×10^{-3}	3.8×10^{-3} **
DeepSVDD [29]	5.25	17/20	6.7×10^{-5}	3.3×10^{-4} ***
KFGOD [31]	5.35	19/20	2.9×10^{-6}	2.0×10^{-5} ***
DisentAD [30]	5.40	17/20	1.6×10^{-4}	6.4×10^{-4} ***
ICL [33]	5.95	18/20	1.3×10^{-5}	8.0×10^{-5} ***
SCMK (ours)	2.23	—	—	—

SCMK against each baseline, applying the Holm correction for multiple comparisons (Table 4). SCMK wins against every baseline on a majority of datasets (13–19 of 20), and all Holm-adjusted p -values are below 0.05 (and below 0.01 for six of the seven), so the improvement is significant in all seven comparisons. The margin is largest over the weaker baselines ($p < 10^{-3}$ for KFGOD, DisentAD, DeepSVDD and ICL) and, while smaller, remains significant against the three top-cluster methods that the Nemenyi test could not separate—LMKAD (14/20, Holm $p = 8.9 \times 10^{-3}$), DFNO (13/20, 1.7×10^{-2}) and NeuTraLAD (16/20, 3.8×10^{-3}).

4.5. Parameter sensitivity

We finally probe how SCMK responds to its three hyper-parameters—the embedding dimension d , the scatter weight λ , and the OC-SVM parameter ν —by varying one at a time on the twenty datasets and averaging the test AUC over the three seeds $\{0, 1, 2\}$. When a parameter is varied, the remaining ones are set to their per-dataset optimum, so each curve reports the AUC attainable at that setting (shaded band: standard deviation across datasets). Fig. 5 summarises the result.

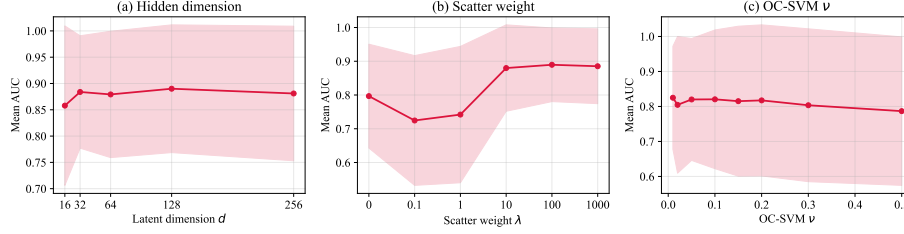


Fig. 5. Parameter sensitivity of SCMK on the twenty datasets (mean test AUC over seeds $\{0, 1, 2\}$; shaded band: standard deviation across datasets). (a) Embedding dimension d and (c) OC-SVM ν are flat (span 0.03–0.04), so SCMK is insensitive to them. (b) The scatter weight λ is the only sensitive parameter: a weak penalty ($\lambda \leq 1$) underperforms even $\lambda = 0$, while $\lambda \geq 10$ is needed for the intra-class compactness that drives detection, plateauing over $[10, 1000]$.

λ is the only sensitive knob. The scatter weight has by far the widest span (mean AUC ranging over 0.165) and a distinctly non-monotone profile: disabling the penalty ($\lambda = 0$) gives 0.797, a *small* weight is actively harmful (0.724 at $\lambda = 0.1$, 0.742 at $\lambda = 1$), and only once $\lambda \geq 10$ does the term pay off, peaking at 0.890 ($\lambda = 100$) and staying flat to $\lambda = 1000$. A weak scatter penalty perturbs the cross-kernel consistency objective without supplying enough intra-class compactness to compensate; a sufficiently strong one delivers the compactness that one-class detection needs. This corroborates the ablation of Section 4.6 and pins the useful range to $\lambda \in [10, 1000]$.

d and ν are robust. By contrast, the embedding dimension is nearly flat (span 0.032): any $d \in \{32, \dots, 256\}$ performs within 0.01 of the $d = 128$ optimum, so SCMK needs no careful width tuning. The OC-SVM ν is likewise stable (span 0.038): AUC is essentially constant for $\nu \leq 0.2$ and declines only gently beyond, favouring a small ν . Overall, SCMK works out-of-the-box for d and ν ; the single design choice that matters is giving the scatter term a large enough weight.

4.6. Ablation study

To isolate the contribution of each component we ablate four design choices on eight representative datasets that deliberately span both detection regimes (angularly separable, e.g. Cardio and Ionosphere; magnitude-separable, e.g. WBC and Autos).

Table 5. Ablation of the four core components (AUC). Best result in each row is in **bold**. The average row shows that the full model is the most robust overall, even though a single-signal variant occasionally peaks on a dataset of its favoured regime.

Dataset	Full	w/o $\mathcal{L}_{\text{scat}}$	w/o $\mathcal{L}_{\text{con}}$	single-kernel	dir. only	mag. only
Ecoli	0.933	0.874	0.931	0.858	0.902	0.969
WBC	0.985	0.981	0.984	0.986	0.586	0.998
Vowels	0.978	0.330	0.977	0.963	0.909	0.906
Cardio	0.967	0.901	0.844	0.703	0.967	0.918
Autos	0.805	0.696	0.802	0.837	0.830	0.763
Ionosphere	1.000	0.866	1.000	1.000	1.000	0.976
Bands-6	0.957	0.517	0.954	0.920	0.967	0.917
Sonar	0.984	0.794	0.978	0.984	0.978	0.951
Average	0.951	0.745	0.934	0.906	0.892	0.925

The variants are: removing the scatter loss (w/o $\mathcal{L}_{\text{scat}}$, i.e. $\lambda = 0$); removing the cross-kernel contrastive loss (w/o $\mathcal{L}_{\text{con}}$); replacing the five-kernel bank with a single kernel (*single-kernel*); and restricting the detector to one signal (*dir. only*, the angular score; *mag. only*, the radial score). Table 5 reports the results, and the average row quantifies the mean impact of each removal.

Scatter loss is the dominant component.. Removing the scatter regularizer ($\lambda = 0$) causes by far the largest degradation, -0.206 on average, collapsing Vowels from 0.978 to 0.330 and Bands-6 from 0.957 to 0.517. This confirms the central thesis of the paper: cross-kernel *consistency* alone does not deliver the intra-class *compactness* that one-class detection requires, and the scatter penalty is what supplies it.

Dual-signal fusion is what makes the detector robust.. The two single-signal variants each excel on one regime but *collapse* on the other. The angular detector (*dir. only*) matches the full model on angularly separable data (Cardio 0.967, Ionosphere 1.000, Sonar 0.978) yet falls to 0.586 on WBC, where normal and malignant cells are nearly indistinguishable in angle; conversely the radial detector (*mag. only*) handles WBC (0.998) but drops to 0.763 on Autos. The fused detector never collapses and attains

the highest mean (0.951) *without* any prior knowledge of which signal is informative. The occasional cases where a single signal slightly exceeds the full model on its own favoured dataset (e.g. *dir. only* on Cardio) are precisely offset by its collapse elsewhere — direct evidence that the fusion, not either signal alone, is the robust choice.

Multi-kernel and contrastive terms. Reducing the bank to a single kernel costs -0.045 on average but is strongly dataset-dependent: it removes 0.26 AUC on Cardio, where multi-scale structure is essential, while leaving datasets with a single dominant scale almost unaffected. Removing the cross-kernel contrastive loss costs the least on average (-0.017); once the scatter penalty is present, the additional consistency constraint contributes only modestly to accuracy, yet it remains important as the uniformity counterweight that prevents representational collapse (Section 3.5).

5. Conclusion and future work

This paper presented SCMK, a scatter-regularized cross-multi-kernel contrastive framework for one-class anomaly detection. SCMK reconciles a bank of multi-scale Gaussian kernels through a cross-kernel InfoNCE objective that treats the kernel index as the contrastive view, and it compacts the normal class in every kernel-induced space through a multi-kernel scatter regularizer that we proved to be equivalent, up to an additive constant, to the average within-class scatter and to the one-class instance of centroid multiple-kernel k -means. The learned representation is exploited by a lightweight dual-signal detector that fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, recovering the discriminative information that ℓ_2 normalization alone discards. Across twenty benchmark datasets and seven recent detectors, SCMK attained the highest mean AUC (0.914) and the best mean rank (2.23), with the improvement confirmed significant by Friedman–Nemenyi and Holm-corrected Wilcoxon tests; ablation and sensitivity studies isolated the scatter term as the dominant component and the dual-signal fusion as the source of robustness, while the runtime analysis showed that the multiple-kernel training premium is bounded and inference stays near-instant.

Several directions remain open. The kernel bank currently uses a fixed median-calibrated Gaussian family; learning the bandwidths, or extending to richer kernel

families, could adapt the multi-resolution prior to each dataset. The present protocol assumes an uncontaminated normal training set, so relaxing SCMK to tolerate label noise or a small anomaly contamination during training is a natural next step. Finally, because inference reduces to a forward pass and two OC-SVM evaluations rather than a transductive recomputation, extending SCMK to streaming and high-dimensional (e.g. image or graph) settings is a promising avenue for scaling the method beyond tabular anomaly detection.

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